

Summary & Findings

Task 1: Data Cleaning and Formatting

There are no null or duplicate values in the data

Task 2: Descriptive Transactional Analysis

1. There is a huge gap in top and bottom performing accounts based on inflow, we can see that the average inflow of top 5 is 260180, where as average inflow of bottom performing account is 7462
2. account has never been completely inactive, these accounts are becoming active and inactive in different time periods, for example if we see ACC10117 it has been inactive 3 times, but it was active as well between (apart from) this times periods, where 37% accounts have been inactive for at least once, and there are 3 account which has been inactive for 5 times

Task 3: Customer Profile Building

1. After categorization of accounts into high, medium and low based on
 - a. Frequency – 45% are with low, 38% mid and 16% are with high frequency
 - b. Average account balance – 49% of accounts has low account balance, 24% has medium and 25% accounts have high account balance, *same is with transaction volume*
2. There are 31 accounts with high net inflow of 67650

3. There are 3 accounts with high frequency of 7 and low average account balance below 59999

4. While average account balance is not zero or below zero for any account, but there are 21 accounts whose account balance was below zero for some certain time period

Task 4: Financial Risk Identification

1. There are 51 transactions with large withdrawal of more than 74047, from 44 accounts, there are 6 accounts with 2 large withdrawals

2. The overall variation of account balance is 48.93, there are 24 accounts with less than 10 coefficients of variance, also 12 accounts are with 0 volatility rate in their account balance, we can see 75% of our account have a coefficient of variation below 50

3. Number of outliers in each numerical variable

a. Transaction amount = 5

b. Account balance = 8

c. Risk score = 8

d. Credit rating = 0

e. Tenure month = 0

Task 5: Visualisation & EDA

1. Analysis of account type

a. Credit accounts = Out of total 27% are credit account with average account balance of 75925 and average transaction amount of 54882

- b. Loan accounts = Out of total 26% are Loan account with average account balance of 71929 and average transaction amount of 56449
- c. Saving accounts = Out of total 25% are Saving account with average account balance of 72015 and average transaction amount of 53282
- d. Current accounts = Out of total 22% are Current account with average account balance of 71542 and average transaction amount of 56387

2. Analysis of transaction type

- a. Deposit = Out of total transactions 25% are deposits with average transaction amount of 54708
- b. withdrawal = Out of total transactions 25% are withdrawal with average transaction amount of 56500
- c. Payment = Out of total transactions 24% are Payment with average transaction amount of 56119
- d. Transfers = Out of total transactions 24% are Transfers with average transaction amount of 53545

3. Analysis of Regional transactions

- a. central region = Out of total transactions 22% transactions are done from central region with average transaction amount of 55677, also average accounts balance of all the customers in central region is 74572
- b. East region = Out of total transactions 18.50% transactions are done from central region with average transaction amount of 56314, also average accounts balance of all the customers in central region is 71400

- c. West region = Out of total transactions 20% transactions are done from central region with average transaction amount of 55132, also average accounts balance of all the customers in central region is 70598
- d. North region = Out of total transactions 19% transactions are done from central region with average transaction amount of 55339, also average accounts balance of all the customers in central region is 68899
- e. South region = Out of total transactions 21% transactions are done from central region with average transaction amount of 53615, also average accounts balance of all the customers in central region is 77676

Task 6: Hypothesis Testing

We have to conduct hypothesis testing to check if accounts with high transaction volume have significantly high average account balance.

Here we have to compare mean of two different groups.

Hence using two-sample t-test

Also checked distribution of account balance , and found it is almost normally distributed

H_0 = mean of high_vol_accounts are not greater than low_vol_accounts (mean of high_val_account=mean of low_val_accounts, or mean of high_val_account < mean of low_val_accounts)

H_a = mean of high_val_accounts > mean of low_val_accounts

significance level = 0.05

after conducting test found

$t_value = 0.7822013389567555$

$p_value = 0.2177154328133169$

P_value is greater than significance level , hence we fail to reject the null value

VIDEO LINK

<https://drive.google.com/file/d/1gCm31OKmcnMNCpSSg6gfiOb5GSLm6MDa/view?usp=sharing>